Intrinsic robust multiscale filtering

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Abstract

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Contents

1 Introduction **2**

2 Intrinsic multiscale filtering **3**

2.1 Local contrast fit 3
2.2 Iterative procedure 3
2.3 Parameter choice and output 4
2.4 True IMFs 4
2.5 Theoretical study for local linear smoothing 4
2.5.1 Linear representation of the IMFs 5
2.5.2 Linear decomposition of the IMFs 6
2.5.3 Noise propagation 6
2.5.4 Regular design 7

3 Robust filtering **7**

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3.1 First-step decomposition 9
3.2 IMFs expansions 10

4 Theoretical guarantees **10**
4.1 Perturbation theory in action 11

A Smooth perturbed optimization **13**
A.1 Gateaux smoothness and self-concordance 13
A.2 Perturbed optimization 15
A.3 Smoothness conditions 16
A.4 A linear perturbation 16
A.5 Quadratic penalization 17
A.6 A generic perturbation 18

B Application **20**

1 Introduction

2 Intrinsic multiscale filtering

Suppose that observed data Y is composed of a signal $X = (X_t)$ and a noise ε_t:

$$Y_t = X_t + \varepsilon_t \,.$$

2.1 Local contrast fit

Fix a contrast function $\rho(\cdot)$, a kernel function $K(\cdot)$, and a bandwidth h. For every t, compute

$$\widetilde{S}_t = \operatorname*{arginf}_x \sum_u \rho(Y_u - x) K_h(t-u) \,,$$

where $K_h(y) = K(y/h)$.

For $\rho(y) = y^2/2$, the value $\widetilde{S}_t$ is the local mean of Y_u for $u \in [t-h, t+h]$:

$$\widetilde{S}_t = \frac{1}{N_h(t)} \sum_u Y_u \, K_h(t-u) \,, \qquad N_h(t) \stackrel{\text{def}}{=} \sum_u K_h(t-u) \,,$$

The choice $\rho(y) = |y|$ leads to the local median.

This step results in the first IMF $\widetilde{S}_t^{(1)} = \widetilde{S}_t$ and the residuals $Y_t^{(2)} = Y_t - \widetilde{S}_t$.

2.2 Iterative procedure

With $Y_t^{(1)} \stackrel{\text{def}}{=} Y_t$, at every step $k \geq 1$, fix h_k and compute

$$\widetilde{S}_t^{(k)} = \operatorname*{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}\big(t-u\big) \,, \quad t \in [0,1] \,.$$

(Here $Y_t^{(k)} = Y_{1+t}^{(k)}$ for all t.) Compute the new residuals

$$Y_t^{(k+1)} = Y_t^{(k)} - \widetilde{S}_t^{(k)} \,.$$

At every step k, decrease h_k by a fixed factor a (e.g. $a = \sqrt{2}$):

$$h_{k+1} = h_k / a \,.$$

Continue until $k \leq k_{\max}$. Alternatively, fix the smallest bandwidth $h_{\min}$ and stop if $h_{k+1} < h_{\min}$.

2.3 Parameter choice and output

Sequence of bandwidths h_k start with $h_1 \asymp 1$ (very big, e.g. 0.1). The starting bandwidth should ensure that each window $[t-h_1, t+h_1]$ contains about half of all observations. This ensures that for the first IMF, almost all noise is smoothed out, and only gross features remain.

Kernel We apply the Epanechnikov kernel:

$$K(u) = \frac{3}{4}(1-|u|)_+^2 .$$

Bandwidth factor The bandwidth factor a can be any value between one and two. Our default choice is $a = \sqrt{2}$ or $a = 2$.

Output The procedure delivers a sequence of IMF's

$$\widetilde{S}^{(1)}, \widetilde{S}^{(2)}, \ldots, \widetilde{S}^{(k_{\max})} .$$

This collection of the IMFs $(\widetilde{S}^{(k)})$ can be used for classification/identification.

2.4 True IMFs

The computed IMFs can be viewed as empirical/noisy versions of their population counterparts (true IMFs) defined with $X_t^{(1)} = X_t$ and the same sequence h_k by

$$S_t^{(k)} = \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u^{(k)} - x) K_{h_k}(t-u) .$$

The question under study is the quality of recovery

$$\|\widetilde{S}^{(k)} - S^{(k)}\|^2 = \sum_{t=1}^n |\widetilde{S}_t^{(k)} - S_t^{(k)}|^2$$

or

$$\sup_t \left|\widetilde{S}_t^{(k)} - S_t^{(k)}\right| .$$

2.5 Theoretical study for local linear smoothing

Fix $\rho(u) = u^2/2$ corresponding to local mean procedure.

2.5.1 Linear representation of the IMFs

Define $W^{(k)}\colon \mathbb{R}^n \to \mathbb{R}^n$ as a linear mapping with the rows

$$W^{(k)} = \Big(\frac{1}{N_k(t)} K_{h_k}(t-u)\Big)_{t,u\in[0,1]} \tag{2.1}$$

with

$$N_k(t) \stackrel{\text{def}}{=} \sum_u K_{h_k}(t-u)\,.$$

Also, define

$$\mathcal{A}^{(k)} \stackrel{\text{def}}{=} (\mathbb{I} - W^{(k)}) \circ \ldots \circ (\mathbb{I} - W^{(1)})$$
$$\mathcal{W}^{(k)} \stackrel{\text{def}}{=} W^{(k)}\mathcal{A}^{(k-1)}\,.$$

Lemma 2.1. *Each* $\widetilde{S}^{(k)} \in \mathbb{R}^n$ *is a linear transform of the residuals* $Y^{(k)} \stackrel{\text{def}}{=} \mathcal{A}^{(k-1)}Y$ *:*

$$\widetilde{S}^{(k)} = W^{(k)}Y^{(k)} = W^{(k)}\mathcal{A}^{(k-1)}Y = \mathcal{W}^{(k)}\,Y\,, \qquad . \tag{2.2}$$

Proof. The identities $\widetilde{S}^{(k)} = W^{(k)}Y^{(k)}$ and $Y^{(k+1)} = Y^{(k)} - \widetilde{S}^{(k)}$ imply

$$Y^{(k+1)} = Y^{(k)} - \widetilde{S}^{(k)} = (\mathbb{I} - W^{(k)})Y^{(k)}\,.$$

By induction, for $k \geq 1$,

$$Y^{(k+1)} = \prod_{j=1}^{k}(\mathbb{I} - W^{(j)})Y = \mathcal{A}^{(k)}Y\,.$$

Similarly,

$$\widetilde{S}^{(k+1)} = W^{(k+1)}\prod_{j=1}^{k}(\mathbb{I} - W^{(j)})Y = W^{(k+1)}\mathcal{A}^{(k)}Y = \mathcal{W}^{(k+1)}\,Y\,,$$

and the assertion follows. □

We can treat $(\mathcal{W}^{(k)})$'s as a family of multiscale operators in $\mathbb{R}^n$ mapping the data Y to a family of noisy IMFs $(\widetilde{S}^{(k)})$.

Lemma 2.2. *For every* $k \geq 1$*, the eigenvalues* $\lambda_j(W^{(k)})$ *of the operator* $W^{(k)}$ *satisfy*

$$\lambda_j(W^{(k)}) \in (0,1)\,, \qquad \|W^{(k)}\| = \lambda_{\max}(W^{(k)}) = 1\,.$$

Furthermore, $\mathcal{A}^{(k)}$ *from (2.2) fulfills* $\|\mathcal{A}^{(k)}\| \leq 1$*.*

Proof. Each matrix $W^{(k)}$ is stochastic is the sense that the elements of each row of $W^{(k)}$ are non-negative and sum to one. The result follows by Gershgorin theorem. □

2.5.2 Linear decomposition of the IMFs

Let $X_t = \mathbb{E}Y_t$, $X^{(1)} = X$.

Lemma 2.3. *For every $k \geq 1$, the true IMF $S^{(k)}$ can be represented as a linear transform of X:*

$$S^{(k)} = \mathcal{W}^{(k)} X = W^{(k)} \circ (\mathbb{I} - W^{(k-1)}) \circ \ldots \circ (\mathbb{I} - W^{(1)}) X \,. \tag{2.3}$$

Let $\boldsymbol{\varepsilon} \stackrel{\mathrm{def}}{=} X - S$ and $\mathbb{E}\boldsymbol{\varepsilon} = 0$. Then for $k \geq 1$,

$$\widetilde{S}^{(k)} - S^{(k)} = W^{(k)} \mathcal{A}^{(k-1)} \boldsymbol{\varepsilon} = \mathcal{W}^{(k)} \boldsymbol{\varepsilon} \,,$$

where $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{(0)} \stackrel{\mathrm{def}}{=} Y - X$ and

$$\begin{aligned} \boldsymbol{\varepsilon}^{(k)} &\stackrel{\mathrm{def}}{=} (\mathbb{I} - W^{(k)}) \boldsymbol{\varepsilon}^{(k-1)} \\ &= (\mathbb{I} - W^{(k)}) \circ (\mathbb{I} - W^{(k-1)}) \circ \ldots \circ (\mathbb{I} - W^{(1)}) \boldsymbol{\varepsilon} = \mathcal{A}^{(k)} \boldsymbol{\varepsilon} \,. \end{aligned} \tag{2.4}$$

Proof. We can use

$$S^{(k)} = W^{(k)} X^{(k)} \,, \qquad X^{(k+1)} = X^{(k)} - S^{(k)} \,.$$

Statement (2.3) follows similarly to (2.2). □

2.5.3 Noise propagation

Let $\boldsymbol{\varepsilon}^{(k)} = \mathcal{A}^{(k)} \boldsymbol{\varepsilon}$ be given by (2.4). Obviously, $\mathbb{E}\boldsymbol{\varepsilon}^{(k)} = 0$. Define $\Sigma^{(k)} = \operatorname{Var}(\boldsymbol{\varepsilon}^{(k)})$.

Lemma 2.4. *It holds for $k \geq 1$*

$$\Sigma^{(k)} = \operatorname{Var}(\boldsymbol{\varepsilon}^{(k)}) = \mathcal{A}^{(k)} \operatorname{Var}(\boldsymbol{\varepsilon}) {\mathcal{A}^{(k)}}^{\top} \,.$$

In particular, if $\operatorname{Var}(\varepsilon_t) = \sigma^2$ then $\Sigma^{(k)} = \mathcal{A}^{(k)} {\mathcal{A}^{(k)}}^{\top}$. Furthermore, $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma)$ yields $\boldsymbol{\varepsilon}^{(k)} \sim \mathcal{N}(0, \Sigma^{(k)})$.

We can use

$$\mathbb{E}\|\widetilde{S}^{(k)} - S^{(k)}\|^2 = \mathbb{E}\|\boldsymbol{\varepsilon}^{(k)}\|^2 = \operatorname{tr} \Sigma^{(k)} \,.$$

Compute and plot $\operatorname{tr} \Sigma^{(k)} = \operatorname{tr}\bigl(\mathcal{A}^{(k)} {\mathcal{A}^{(k)}}^{\top}\bigr)$ for $k = 1, \ldots, k_{\max}$.

Lemma 2.5. *It holds $\bigl(W^{(k)} {W^{(k)}}^{\top}\bigr)_{tt} \asymp (nh_k)^{-1}$.*

2.5.4 Regular design

Assume a regular deterministic design $t = 1/n, 2/n, \ldots, 1$ and the vector $Y = (Y_t)$ is extended to $t \in [-0.5, 1.5]$; e.g. by $Y_t = Y_{1+t}$ for $t < 0$.

Lemma 2.6. *Assume a regular design with $N_k(t) \equiv N_k$ for $t \in [-0.5, 1.5]$. Then the operators $W^{(k)}$ from (2.1) and $\mathcal{A}^{(k)}$ from (2.2) mutually commute.*

Proof. Fix $k', k \geq 1$. Then for $t \in [0, 1]$

$$\begin{aligned}(W^{(k)}W^{(k')}Y)_t &= \sum_u \frac{K_{h_k}(t-u)}{N_k(t)}(W^{(k')}Y)_u \\ &= \sum_u \sum_s \frac{K_{h_k}(t-u)}{N_k(t)} \frac{K_{h_{k'}}(u-s)}{N_{k'}(u)} Y_s \\ &= \frac{1}{N_k N_{k'}} \sum_s \biggl\{ \sum_u K_{h_k}(t-u) K_{h_{k'}}(u-s) \biggr\} Y_s \,.\end{aligned}$$

The same representation applies to $(W^{(k')}W^{(k)}Y)_t$. The statement about $\mathcal{A}^{(k)}$ follows from (2.2). □

3 Robust filtering

This section presents the robustified procedure based on local median filtering. More precisely, we apply a smoothed version of median filtering. Huber's proposal leads to

$$\rho(x) \stackrel{\text{def}}{=} \tfrac{1}{2}x^2 \, \mathbb{1}(|x| \leq H) + (Hx - \tfrac{1}{2}H^2) \, \mathbb{1}(|x| > H) \,.$$

The value H is a tuning meta-parameter. Our theoretical results require 3 derivatives of ρ. We consider the third-order spline with $\rho(0) = \rho'(0) = 0$, $\rho''(0) = 1$, and

$$\rho'''(x) = -\operatorname{sign}(x) \, \mathbb{1}(H \leq |x| \leq 2H) \,.$$

The function $\rho(x)$ is quadratic for $|x| \leq H$ with $\rho(x) = x^2/2$, then the second derivative $\rho(x)$ decreases from one to zero at $x = 2H$. As $\rho''(x) \geq 0$, this function is convex.

Alternatively, with $\rho_0(x) = |x|$ and $\varphi_H(x) = \frac{1}{\sqrt{2\pi}H} \exp\Bigl(-\frac{x^2}{2H^2}\Bigr)$, one can use

$$\rho_H(x) \stackrel{\text{def}}{=} \rho_0 * \varphi_H(x) \,.$$

This value can be computed analytically.

Lemma 3.1. *It holds*

$$\rho_H(x) = x\{2\Phi(x/H) - 1\} + \frac{2H}{\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr)\,. \tag{3.1}$$

Also, for $x \geq 0$,

$$\begin{aligned}
\rho'_H(x) &= 2\Phi(x/H) - 1\,,\\
\rho''_H(x) &= \frac{2}{H}\phi(x/H) = \frac{2}{H\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr)\,,\\
\rho'''_H(x) &= -\frac{2x}{H^3\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr)\,,
\end{aligned}$$

Proof. We use

$$\rho_H(x) = \int_{-\infty}^{\infty} |t|\, \varphi_H(x-t)dt = H\int_{-\infty}^{\infty} |t|\, \varphi(x/H - t)dt = H\int_{-\infty}^{\infty} |t + x/H|\, \varphi(t)dt\,.$$

For any $a \geq 0$,

$$\begin{aligned}
&\int_{-\infty}^{\infty} |t+a|\, \varphi(t)dt = \frac{1}{2}\int_{-\infty}^{\infty} \bigl(|t+a| + |t-a|\bigr)\, \varphi(t)dt = a\int_{-a}^{a} \varphi(t)dt + 2\int_{a}^{\infty} t\, \varphi(t)dt\\
&\quad = a\bigl\{\Phi(a) - \Phi(-a)\bigr\} + \frac{2}{\sqrt{2\pi}}\int_{a}^{\infty} \mathrm{e}^{-t^2/2} d(t^2/2) = a\bigl\{\Phi(a) - \Phi(-a)\bigr\} + \frac{2}{\sqrt{2\pi}}\mathrm{e}^{-a^2/2}\,.
\end{aligned}$$

This implies (3.1). The other identities follow by direct differentiation. □

Lemma 3.2. *Check that* $\rho_H(x)$ *is*

- *infinitely differentiable;*
- *symmetric, convex, positive, and monotonously increasing with* $|x|$*;*
- *it holds for all* x

$$\begin{gathered}
|\rho'_H(x)| \leq 1\,, \qquad 1 - |\rho'_H(x)| \leq 2\bigl\{1 - \Phi(|x|/H)\bigr\}\,,\\
0 \leq \rho''_H(x) \leq \frac{2}{H\sqrt{2\pi}}; \qquad |\rho'''_H(x)| \leq \frac{2}{H^2\sqrt{2\pi}}; \qquad \bigl|\rho'''_H(x)\bigr| \leq \frac{|x|}{H^2}\rho''_H(x)\,.
\end{gathered}$$

- *for* $|x| > 3H$, $\bigl|\rho_H(x) - |x|\bigr|$ *is very small:*

$$\bigl|\rho_H(x) - |x|\bigr| \leq \frac{2H}{\sqrt{2\pi}} \exp\bigl(-\frac{x^2}{2H^2}\bigr) \leq 0.009H\,, \qquad |x| > 3H\,.$$

The procedure reads as follows:

- Set $Y^{(1)} \equiv Y$, $k = 1$;

- compute

$$\begin{aligned} \widetilde{S}_t^{(k)} &= \operatorname*{arginf}_x \sum_u \rho(Y_u^{(k)} - x) K_{h_k}(t-u)\,; \\ Y_t^{(k+1)} &= Y_t^{(k)} - \widetilde{S}_t^{(k)}\,; \end{aligned} \tag{3.2}$$

- recompute $h_{k+1} = h_k/a\,;$

- increase k by 1; loop until $k > k_{\max}$ or $h_k < h_{\min}\,;$

Choice of H is important. This value should be of the same order as the standard deviation of the noise before contamination.

3.1 First-step decomposition

For a general ρ-function, one of the main questions is the definition of the true IMF. Define

$$S_t \stackrel{\text{def}}{=} \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u - x) K_h(t-u)\,.$$

If $Y_t = X_t + \varepsilon_t$ for ε_t i.i.d. then

$$\mathbb{E}\rho(Y_u - x) = \phi(X_u - x)\,, \quad \phi(x) \stackrel{\text{def}}{=} \mathbb{E}\rho(x+\varepsilon)\,,$$

leading to

$$S_t = \operatorname*{arginf}_x \sum_u \phi(X_u - x) K_h(t-u)\,.$$

Lemma 3.3. *ρ convex implies ϕ convex.*

Define

$$N_t(x) \stackrel{\text{def}}{=} \sum_u \mathbb{E}\rho''(Y_u - x) K_h(t-u) = \sum_u \phi''(X_u - x) K_h(t-u) \tag{3.3}$$

and

$$N_t \stackrel{\text{def}}{=} N_t(X_t)\,.$$

Also, denote

$$\diamondsuit_{u,t} \stackrel{\text{def}}{=} \rho'(Y_u - X_t) - \mathbb{E}\rho'(Y_u - X_t) = \rho'(Y_u - X_t) - \phi'(X_u - X_t)\,.$$

Obviously, $\mathbb{E}\diamondsuit_{u,t} \equiv 0$. The following expansion will be a key tool of analysis:

$$\widetilde{S}_t - S_t \approx \frac{1}{N_t} \sum_u \diamondsuit_{u,t} K_h(t-u) = \sum_u \diamondsuit_{u,t}\, w_{u,t}\,,$$
$$w_{u,t} \stackrel{\text{def}}{=} \frac{K_h(t-u)}{N_t}\,.$$

3.2 IMFs expansions

Set $Y^{(1)} \equiv Y$ and $X^{(1)} \equiv X$.

Step $k \geq 1$: with $\varepsilon_u^{(k)} \stackrel{\text{def}}{=} Y_u^{(k)} - X_u^{(k)}$

$$\begin{aligned} S_t^{(k)} &= \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u^{(k)} - x) K_h(t-u) \\ &= \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(X_u^{(k)} + \varepsilon_u^{(k)} - x) K_h(t-u) \\ &= \operatorname*{arginf}_x \sum_u \mathbb{E}\phi_k(X_u^{(k)} - x, u) K_h(t-u)\,, \end{aligned}$$

where

$$\phi_k(x,u) \stackrel{\text{def}}{=} \mathbb{E}\rho(x + \varepsilon_u^{(k)})\,.$$

Continue with $Y^{(k+1)} = Y^{(k)} - \widetilde{S}^{(k)}$ from (3.2) and

$$X^{(k+1)} = X^{(k)} - S^{(k)}\,.$$

4 Theoretical guarantees

This section describes the accuracy of the procedure. The starting point is an accuracy bounds for $\widetilde{S}_t - S_t$ for a fixed $t \in [0,1]$. We use tools from perturbed optimization; see Section A. Represent

$$\begin{aligned} \widetilde{S}_t &= \operatorname*{arginf}_x \sum_u \rho(Y_u - x) K_h(t-u) \\ S_t &= \operatorname*{arginf}_x \sum_u \mathbb{E}\rho(Y_u - x)\, K_h(t-u) = \operatorname*{arginf}_x \sum_u \phi(X_u - x) K_h(t-u)\,. \end{aligned} \tag{4.1}$$

Also, define

$$\mathbb{F}_t \stackrel{\text{def}}{=} \sum_u \phi''(X_u - S_t) K_h(t-u)\,,$$
$$\nabla_t \stackrel{\text{def}}{=} \sum_u \bigl\{\rho'(Y_u - S_t) - \mathbb{E}\rho'(Y_u - S_t)\bigr\} K_h(t-u)\,,$$

and $D_t \stackrel{\text{def}}{=} \mathbb{F}_t^{1/2}$. Note that $\mathbb{F} = N_t$; see (3.3).

Proposition 4.1. *On a random set* $\Omega(\mathtt{x})$ *with* $\mathbb{P}\bigl(\Omega(\mathtt{x})\bigr) \geq 1 - \mathrm{e}^{-\mathtt{x}}$

$$\sup_{t\in[0,1]} \bigl\| D_t(\widetilde{S}_t - S_t + \mathbb{F}_t^{-1}\nabla_t) \bigr\| \leq \quad , \qquad .$$

4.1 Perturbation theory in action

Consider two optimization problems in (4.1) using the perturbed optimization approach. We fix $t \in [0,1]$. All the quantities considered below will depend on t. Define

$$\begin{aligned} f(x) &\stackrel{\text{def}}{=} \sum_u \phi(X_u - x) K_h(t-u)\,, \\ f_\circ(x) &\stackrel{\text{def}}{=} \sum_u \rho(Y_u - x) K_h(t-u)\,. \end{aligned} \tag{4.2}$$

The aim is to compare the solutions

$$x^* \stackrel{\text{def}}{=} \operatorname*{arginf}_x f(x)\,, \qquad x_\circ \stackrel{\text{def}}{=} \operatorname*{arginf}_x f_\circ(x)\,.$$

Unfortunately, the perturbation $\zeta(x) \stackrel{\text{def}}{=} f(x) - f_\circ(x)$ is not linear in x. We apply Theorem **??** to f and $f_\circ$ from (4.2). The functions $f(x)$ and $f_\circ(x)$ are strongly convex in x. Define

$$\begin{aligned} \mathbb{F}_\circ &\stackrel{\text{def}}{=} f_\circ''(x^*) = \sum_u \rho''(Y_u - x^*) K_h(t-u)\,, \\ \mathbb{F} &\stackrel{\text{def}}{=} f''(x^*) = \sum_u \phi''(X_u - x^*) K_h(t-u)\,. \end{aligned}$$

Also, define $D = \mathbb{F}^{1/2}$.

Lemma 4.2. *It holds on* $\Omega(\mathtt{x})$

(i)

$$\begin{aligned} |f'(x^*) - f_\circ'(x^*)| &= \frac{1}{D}\Bigl|\sum_u \bigl\{\rho'(Y_u - x^*) - \mathbb{E}\rho'(Y_u - x^*)\bigr\} K_h(t-u)\Bigr| \leq \mathtt{C}_1\,, \\ |f''(x^*) - f_\circ''(x^*)| &= \frac{1}{D}\Bigl|\sum_u \bigl\{\rho''(Y_u - x^*) - \mathbb{E}\rho''(Y_u - x^*)\bigr\} K_h(t-u)\Bigr| \leq \mathtt{C}_2\,, \end{aligned}$$

(ii)

$$\sup_{x\in U} \frac{1}{D^2}\Bigl|\sum_u \rho'''(Y_u - x) K_h(t-u)\Bigr| \leq \mathtt{C}_3\,,$$

(iii)

$$\sup_{x\in U}\frac{1}{D^2}\Big|\sum_u \phi''(X_u - x) - \phi''(X_u - x^*)K_h(t-u)\Big| \le$$

Proof. This is a standard Bernstein-type deviation bound for independent sums under Lipschitz condition on each summand; see e.g. Boucheron et al. (2003). □

Lemma 4.2 (ii) implies

$$\frac{1}{\mathbb{F}}\big|\mathbb{F}_\circ - \mathbb{F}\big| \le \delta \stackrel{\text{def}}{=} \frac{\mathtt{C}_2}{D}\,. \tag{4.3}$$

Further, consider

$$\mathtt{b}_\circ \stackrel{\text{def}}{=} \frac{D}{\mathbb{F}_\circ}|f'_\circ(x^*)| = \frac{D}{\mathbb{F}_\circ}\Big|\sum_u \rho'(Y_u - x^*)K_h(t-u)\Big|\,.$$

As $f'(x^*) = 0$, it also holds

$$\mathtt{b}_\circ = \frac{D}{\mathbb{F}_\circ}\Big|\sum_u \big\{\rho'(Y_u - x^*) - \mathbb{E}\rho'(Y_u - x^*)\big\}K_h(t-u)\Big|\,.$$

Lemma 4.2 (i) and (4.3) imply

$$\mathtt{b}_\circ \le \frac{\mathtt{C}_1}{1-\delta}\,.$$

Then, by (A.7) of Theorem **??** with $\tau_3 = \mathtt{C}_3/D$,

$$\big|D^{-1}\mathbb{F}_\circ(x_\circ - x^* + \mathbb{F}_\circ^{-1}f'_\circ(x^*))\big| \le \frac{3\mathtt{C}_3}{4D}\,\mathtt{b}_\circ^2 \tag{4.4}$$

yielding by (4.3)

$$\big|x_\circ - x^* + \mathbb{F}_\circ^{-1}f'_\circ(x^*)\big| \le \frac{3\mathtt{C}_3}{4D^2(1-\delta)}\,\mathtt{b}_\circ^2 \le \frac{3\mathtt{C}_1^2\,\mathtt{C}_3}{4D^2(1-\delta)^3}\,.$$

Also,

$$\big|\mathbb{F}_\circ^{-1}f'_\circ(x^*) - \mathbb{F}^{-1}f'_\circ(x^*)\big| = \frac{|\mathbb{F}_\circ - \mathbb{F}|}{\mathbb{F}_\circ\mathbb{F}}\,|f'_\circ(x^*)| \le \frac{\delta}{1-\delta}\,\frac{1}{\mathbb{F}}\,|f'_\circ(x^*)| \le \frac{\mathtt{C}_1\,\mathtt{C}_2}{D^2(1-\delta)}$$

Together with

Approximation (4.3) enables us to replace $\mathbb{F}_\circ$ with $\mathbb{F}$ everywhere and to rewrite the bound (4.4) as

$$\big|x_\circ - x^* + \mathbb{F}^{-1}f'_\circ(x^*)\big| \le \frac{\mathtt{C}}{D^2}\,.$$

A Smooth perturbed optimization

This section presents conditions and accurate bounds for smooth perturbed optimization. We consider minimization of a smooth convex function f and study the impact of a linear, quadratic, or smooth perturbation.

A.1 Gateaux smoothness and self-concordance

Below we assume the function $f(\upsilon)\,,\ \upsilon \in \Upsilon \subseteq \mathbb{R}^p$ to be strongly convex with a positive definite Hessian $\mathbb{F}(\upsilon) \stackrel{\text{def}}{=} \nabla^2 f(\upsilon)\,.$ Also, assume $f(\upsilon)$ three or sometimes even four times Gateaux differentiable in $\upsilon \in \Upsilon\,.$ Local smoothness of f will be described by the relative error of the Taylor expansion of the third or fourth order. Define

$$\begin{aligned}
\delta_3(\upsilon, u) &= f(\upsilon + u) - f(\upsilon) - \langle \nabla f(\upsilon), u \rangle - \frac{1}{2}\langle \nabla^2 f(\upsilon), u^{\otimes 2} \rangle, \\
\delta_3'(\upsilon, u) &= \langle \nabla f(\upsilon + u), u \rangle - \langle \nabla f(\upsilon), u \rangle - \langle \nabla^2 f(\upsilon), u^{\otimes 2} \rangle\,,
\end{aligned}$$

and

$$\delta_4(\upsilon, u) = f(\upsilon + u) - f(\upsilon) - \langle \nabla f(\upsilon), u \rangle - \frac{1}{2}\langle \nabla^2 f(\upsilon), u^{\otimes 2} \rangle - \frac{1}{6}\langle \nabla^3 f(\upsilon), u^{\otimes 3} \rangle\,.$$

Now, for each $\upsilon\,,$ suppose to be given a positive symmetric matrix $D(\upsilon)$ defining a local metric and a local vicinity around $\upsilon\colon$ for some radius $\mathtt{r}\,,$

$$\mathcal{U}_{\mathtt{r}}(\upsilon) = \bigl\{ u \in \mathbb{R}^p \colon \| D(\upsilon) u \| \le \mathtt{r} \bigr\}$$

Local smoothness properties of f at υ are given via the quantities

$$\omega(\upsilon) \stackrel{\text{def}}{=} \sup_{u\colon \|D(\upsilon)u\| \le \mathtt{r}} \frac{2|\delta_3(\upsilon, u)|}{\|D(\upsilon)u\|^2}\,, \quad \omega'(\upsilon) \stackrel{\text{def}}{=} \sup_{u\colon \|D(\upsilon)u\| \le \mathtt{r}} \frac{|\delta_3'(\upsilon, u)|}{\|D(\upsilon)u\|^2}\,. \tag{A.1}$$

The definition yields for any u with $\|D(\upsilon)u\| \le \mathtt{r}$

$$\bigl|\delta_3(\upsilon, u)\bigr| \le \frac{\omega(\upsilon)}{2}\|D(\upsilon)u\|^2\,, \qquad \bigl|\delta_3'(\upsilon, u)\bigr| \le \omega'(\upsilon)\|D(\upsilon)u\|^2\,. \tag{A.2}$$

The approximation results can be improved under a third-order upper bound on the error of Taylor expansion.

$(\mathcal{T}_3)$ *For some* τ_3

$$\bigl|\delta_3(\upsilon, u)\bigr| \le \frac{\tau_3}{6}\|D(\upsilon)\,u\|^3\,, \quad \bigl|\delta_3'(\upsilon, u)\bigr| \le \frac{\tau_3}{2}\|D(\upsilon)\,u\|^3\,, \quad u \in \mathcal{U}_{\mathtt{r}}(\upsilon).$$

$(\mathcal{T}_4)$ *For some* τ_4

$$\bigl|\delta_4(\upsilon,\mathbf{u})\bigr| \le \frac{\tau_4}{24}\|D(\upsilon)\,\mathbf{u}\|^4\,, \qquad \mathbf{u}\in\mathcal{U}_{\mathtt{r}}(\upsilon).$$

We also present a version of $(\mathcal{T}_3)$ resp. $(\mathcal{T}_4)$ in terms of the third (resp. fourth) derivative of f.

$(\mathcal{T}_3^*)$ $f(\upsilon)$ *is three times differentiable and*

$$\sup_{\mathbf{u}:\,\|D(\upsilon)\mathbf{u}\|\le\mathtt{r}}\ \sup_{z\in\mathbb{R}^p}\ \frac{\bigl|\langle\nabla^3 f(\upsilon+\mathbf{u}),z^{\otimes 3}\rangle\bigr|}{\|D(\upsilon)z\|^3} \le \tau_3\,.$$

$(\mathcal{T}_4^*)$ $f(\upsilon)$ *is four times differentiable and*

$$\sup_{\mathbf{u}:\,\|D(\upsilon)\mathbf{u}\|\le\mathtt{r}}\ \sup_{z\in\mathbb{R}^p}\ \frac{\bigl|\langle\nabla^4 f(\upsilon+\mathbf{u}),z^{\otimes 4}\rangle\bigr|}{\|D(\upsilon)z\|^4} \le \tau_4\,.$$

By Banach's characterization Banach (1938), $(\mathcal{T}_3^*)$ implies

$$\bigl|\langle\nabla^3 f(\upsilon+\mathbf{u}),z_1\otimes z_2\otimes z_3\rangle\bigr| \le \tau_3\|D(\upsilon)z_1\|\,\|D(\upsilon)z_2\|\,\|D(\upsilon)z_3\|$$

for any $\mathbf{u}$ with $\|D(\upsilon)\mathbf{u}\|\le\mathtt{r}$ and all $z_1,z_2,z_3\in\mathbb{R}^p$. Similarly under $(\mathcal{T}_4^*)$

$$\begin{aligned}&\bigl|\langle\nabla^4 f(\upsilon+\mathbf{u}),z_1\otimes z_2\otimes z_3\otimes z_4\rangle\bigr|\\ &\quad\le \tau_4\|D(\upsilon)z_1\|\,\|D(\upsilon)z_2\|\,\|D(\upsilon)z_3\|\,\|D(\upsilon)z_4\|\,, \quad z_1,z_2,z_3,z_4\in\mathbb{R}^p\,.\end{aligned}$$

Lemma A.1. *Under* $(\mathcal{T}_3)$ *or* $(\mathcal{T}_3^*)$, *it holds for* $\omega(\upsilon)$ *and* $\omega'(\upsilon)$ *from* (A.1)

$$\omega(\upsilon)\le\frac{\tau_3\,\mathtt{r}}{3}\,, \qquad \omega'(\upsilon)\le\frac{\tau_3\,\mathtt{r}}{2}\,.$$

Proof. For any $\mathbf{u}\in\mathcal{U}_{\mathtt{r}}(\upsilon)$ with $\|D(\upsilon)\mathbf{u}\|\le\mathtt{r}$

$$\bigl|\delta_3(\upsilon,\mathbf{u})\bigr| \le \frac{\tau_3}{6}\,\|D(\upsilon)\mathbf{u}\|^3 \le \frac{\tau_3\,\mathtt{r}}{6}\,\|D(\upsilon)\mathbf{u}\|^2,$$

and the bound for $\omega(\upsilon)$ follows. The proof for $\omega'(\upsilon)$ is similar. Under $(\mathcal{T}_3^*)$, apply the third order Taylor expansion to the univariate function $f(\upsilon+t\mathbf{u})$ of t and $(\mathcal{T}_3^*)$ with $z\equiv\mathbf{u}$. □

The values τ_3 and τ_4 are usually very small. Some quantitative bounds are given later in this section under the assumption that the function $f(\upsilon)$ can be written in the form $f(\upsilon)=nh(\upsilon)$ for a fixed smooth function $h(\upsilon)$ with the Hessian $\nabla^2 h(\upsilon)$. The factor n has meaning of the sample size; see Section **??** or Section **??**.

$(\mathcal{S}_3^*)$ $f(\upsilon) = nh(\upsilon)$ *for* $h(\upsilon)$ *three times differentiable and for a metric tensor* $\boldsymbol{g}(\upsilon)$

$$\sup_{\boldsymbol{u}:\, \|\boldsymbol{g}(\upsilon)\boldsymbol{u}\| \le \mathtt{r}/\sqrt{n}} \frac{\bigl|\langle \nabla^3 h(\upsilon+\boldsymbol{u}), \boldsymbol{u}^{\otimes 3}\rangle\bigr|}{\|\boldsymbol{g}(\upsilon)\boldsymbol{u}\|^3} \le \mathtt{c}_3 \,.$$

$(\mathcal{S}_4^*)$ *the function* $h(\cdot)$ *satisfies* $(\mathcal{S}_3^*)$ *and*

$$\sup_{\boldsymbol{u}:\, \|\boldsymbol{g}(\upsilon)\boldsymbol{u}\| \le \mathtt{r}/\sqrt{n}} \frac{\bigl|\langle \nabla^4 h(\upsilon+\boldsymbol{u}), \boldsymbol{u}^{\otimes 4}\rangle\bigr|}{\|\boldsymbol{g}(\upsilon)\boldsymbol{u}\|^4} \le \mathtt{c}_4 \,.$$

$(\mathcal{S}_3^*)$ and $(\mathcal{S}_4^*)$ are local versions of the so-called self-concordance condition; see Nesterov and Nemirovskii (1994) and Ostrovskii and Bach (2021). In fact, they require that each univariate function $h(\upsilon + t\boldsymbol{u})$ of $t \in \mathbb{R}$ is self-concordant with some universal constants $\mathtt{c}_3$ and $\mathtt{c}_4$. Under $(\mathcal{S}_3^*)$ and $(\mathcal{S}_4^*)$, with $D^2(\upsilon) = n\,\boldsymbol{g}^2(\upsilon)$, the values $\delta_3(\upsilon, \boldsymbol{u})$, $\delta_4(\upsilon, \boldsymbol{u})$, and $\omega(\upsilon)$, $\omega'(\upsilon)$ can be well bounded.

Lemma A.2. *Suppose* $(\mathcal{S}_3^*)$. *Then* $(\mathcal{T}_3)$ *follows with* $\tau_3 = \mathtt{c}_3 n^{-1/2}$. *Moreover, for* $\omega(\upsilon)$ *and* $\omega'(\upsilon)$ *from* (A.1), *it holds*

$$\omega(\upsilon) \le \frac{\mathtt{c}_3\,\mathtt{r}}{3n^{1/2}}\,, \qquad \omega'(\upsilon) \le \frac{\mathtt{c}_3\,\mathtt{r}}{2n^{1/2}}\,. \tag{A.3}$$

Also $(\mathcal{T}_4)$ *follows from* $(\mathcal{S}_4^*)$ *with* $\tau_4 = \mathtt{c}_4 n^{-1}$.

Proof. For any $\boldsymbol{u} \in \mathcal{U}_{\mathtt{r}}(\upsilon)$, by third order Taylor expansion,

$$\begin{aligned} |\delta_3(\upsilon, \boldsymbol{u})| &\le \sup_{t\in[0,1]} \frac{1}{6}\bigl|\langle \nabla^3 f(\upsilon + t\boldsymbol{u}), \boldsymbol{u}^{\otimes 3}\rangle\bigr| = \frac{n}{6} \sup_{t\in[0,1]} \bigl|\langle \nabla^3 h(\upsilon + t\boldsymbol{u}), \boldsymbol{u}^{\otimes 3}\rangle\bigr| \\ &\le \frac{n\,\mathtt{c}_3}{6}\,\|\boldsymbol{g}(\upsilon)\boldsymbol{u}\|^3 = \frac{n^{-1/2}\,\mathtt{c}_3}{6}\,\|D(\upsilon)\boldsymbol{u}\|^3 \le \frac{n^{-1/2}\,\mathtt{c}_3\,\mathtt{r}}{6}\,\|D(\upsilon)\boldsymbol{u}\|^2 \,. \end{aligned}$$

This implies $(\mathcal{T}_3)$ as well as (A.3); see (A.2). The statement about $(\mathcal{T}_4)$ is similar. □

A.2 Perturbed optimization

This section provides an overview of some results from Spokoiny (2025). Let $f(\upsilon)$ be a smooth strongly convex function,

$$\upsilon^* = \operatorname*{arginf}_{\upsilon} f(\upsilon),$$

and $\mathbb{F} = \nabla^2 f(\upsilon^*)$. We study the question of how the point of minimum and the value of minimum of f change under some perturbation of f. More precisely, with $f_\circ(\upsilon)$

being a perturbed version of $f(\boldsymbol{\upsilon})$, we consider in parallel two optimization problems

$$\boldsymbol{\upsilon}^* = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon}), \quad \boldsymbol{\upsilon}_\circ = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f_\circ(\boldsymbol{\upsilon}).$$

The goal is to describe the discrepancies $\boldsymbol{\upsilon}_\circ - \boldsymbol{\upsilon}^*$ and $f_\circ(\boldsymbol{\upsilon}_\circ) - f_\circ(\boldsymbol{\upsilon}^*)$ induced by this perturbation. We also present the results for two specific cases: a linear and a quadratic perturbation.

A.3 Smoothness conditions

Assume $f(\boldsymbol{\upsilon})$ to be convex and three or sometimes even four times Gateaux differentiable in $\boldsymbol{\upsilon} \in \Upsilon$. For a fixed $\boldsymbol{\upsilon} \in \Upsilon$, we need such a condition uniformly in a local vicinity of a fixed point $\boldsymbol{\upsilon} \in \Upsilon$. The notion of locality is given in terms of a metric tensor $D(\boldsymbol{\upsilon}) \in \mathfrak{M}_p$, where $\mathfrak{M}_p$ is the set of symmetric positive p-matrices. In most of the results later on, one can use $D(\boldsymbol{\upsilon}) = \mathbb{F}^{1/2}(\boldsymbol{\upsilon})$, where $\mathbb{F}(\boldsymbol{\upsilon}) = \nabla^2 f(\boldsymbol{\upsilon})$. Introduce the following conditions.

$(\boldsymbol{\mathcal{T}}_3^*)$ *$f(\boldsymbol{\upsilon}+\boldsymbol{u})$ is three times differentiable in $\boldsymbol{u}$ and with D possibly depending on $\boldsymbol{\upsilon}$*

$$\sup_{\boldsymbol{u}:\, \|D\boldsymbol{u}\| \le \mathtt{r}} \sup_{\boldsymbol{z} \in \mathbb{R}^p} \frac{\bigl|\langle \nabla^3 f(\boldsymbol{\upsilon}+\boldsymbol{u}), \boldsymbol{z}^{\otimes 3} \rangle\bigr|}{\|D\boldsymbol{z}\|^3} \le \tau_3 .$$

$(\boldsymbol{\mathcal{T}}_4^*)$ *$f(\boldsymbol{\upsilon}+\boldsymbol{u})$ is four times differentiable in $\boldsymbol{u}$ and*

$$\sup_{\boldsymbol{u}:\, \|D\boldsymbol{u}\| \le \mathtt{r}} \sup_{\boldsymbol{z} \in \mathbb{R}^p} \frac{\bigl|\langle \nabla^4 f(\boldsymbol{\upsilon}+\boldsymbol{u}), \boldsymbol{z}^{\otimes 4} \rangle\bigr|}{\|D\boldsymbol{z}\|^4} \le \tau_4 .$$

By Banach's characterization Banach (1938), $(\boldsymbol{\mathcal{T}}_3^*)$ implies

$$\bigl|\langle \nabla^3 f(\boldsymbol{\upsilon}+\boldsymbol{u}), \boldsymbol{z}_1 \otimes \boldsymbol{z}_2 \otimes \boldsymbol{z}_3 \rangle\bigr| \le \tau_3 \|D\boldsymbol{z}_1\| \, \|D\boldsymbol{z}_2\| \, \|D\boldsymbol{z}_3\|$$

for any $\boldsymbol{u}$ with $\|D\boldsymbol{u}\| \le \mathtt{r}$ and all $\boldsymbol{z}_1, \boldsymbol{z}_2, \boldsymbol{z}_3 \in \mathbb{R}^p$.

The value τ_3 is usually very small. If the function $f(\boldsymbol{\upsilon})$ can be written in the form $f(\boldsymbol{\upsilon}) = nh(\boldsymbol{\upsilon})$ for a fixed smooth function $h(\boldsymbol{\upsilon})$ with the Hessian $\nabla^2 h(\boldsymbol{\upsilon})$, and a factor n meaning of the sample size, then $\tau_3 \asymp n^{-1/2}$. $(\boldsymbol{\mathcal{T}}_3^*)$ is a local version of the so-called self-concordance condition; see Nesterov (1988) and Ostrovskii and Bach (2021).

A.4 A linear perturbation

For the objective function f, let another function $f_\circ(\boldsymbol{\upsilon})$ satisfy for some vector $\boldsymbol{A}$

$$f_\circ(\boldsymbol{\upsilon}) - f_\circ(\boldsymbol{\upsilon}^*) = \bigl\langle \boldsymbol{\upsilon} - \boldsymbol{\upsilon}^*, \boldsymbol{A} \bigr\rangle + f(\boldsymbol{\upsilon}) - f(\boldsymbol{\upsilon}^*). \tag{A.4}$$

Define

$$\boldsymbol{\upsilon}_\circ \stackrel{\text{def}}{=} \operatorname*{arginf}_{\boldsymbol{\upsilon}} f_\circ(\boldsymbol{\upsilon}), \qquad f_\circ(\boldsymbol{\upsilon}_\circ) = \inf_{\boldsymbol{\upsilon}} f_\circ(\boldsymbol{\upsilon}). \tag{A.5}$$

The analysis aims to evaluate the quantities $\boldsymbol{\upsilon}_\circ - \boldsymbol{\upsilon}^*$ and $f_\circ(\boldsymbol{\upsilon}_\circ) - f_\circ(\boldsymbol{\upsilon}^*)$.

Theorem A.3. *Let*

- $f(\boldsymbol{\upsilon})$ *be a strongly convex function with* $f(\boldsymbol{\upsilon}^*) = \inf_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon})$ *and* $\mathbb{F} = \nabla^2 f(\boldsymbol{\upsilon}^*)$;
- $f_\circ(\boldsymbol{\upsilon})$ *and* $f(\boldsymbol{\upsilon})$ *be related by* (A.4) *with some vector* $\boldsymbol{A}$;
- *the function* $f(\boldsymbol{\upsilon})$ *satisfy* $(\boldsymbol{\mathcal{T}}_3^*)$ *at* $\boldsymbol{\upsilon}^*$ *with* D^2, $\mathtt{r} = \mathtt{r}(\boldsymbol{\upsilon}^*)$, *and* τ_3;
- *for some* $\varkappa > 0$

$$D^2 \le \varkappa^2 \mathbb{F}, \quad \mathtt{r} \ge \frac{4}{3}\|D\,\mathbb{F}^{-1}\boldsymbol{A}\|, \quad \varkappa^2 \tau_3 \,\|D\,\mathbb{F}^{-1}\boldsymbol{A}\| \le \frac{3}{8}.$$

Define

$$\rho \stackrel{\text{def}}{=} \varkappa^2 \tau_3 \,\|D\,\mathbb{F}^{-1}\boldsymbol{A}\|, \qquad \ell \stackrel{\text{def}}{=} \frac{2}{1+\sqrt{1-2\rho}}.$$

Then $\ell \in [1, 4/3]$ *and the following bounds are valid:*

$$\begin{aligned} \|D(\boldsymbol{\upsilon}_\circ - \boldsymbol{\upsilon}^*)\| &\le \quad \ell\,\|D\,\mathbb{F}^{-1}\boldsymbol{A}\| \quad \le \quad \frac{4}{3}\,\|D\,\mathbb{F}^{-1}\boldsymbol{A}\|, \\ \|D^{-1}\mathbb{F}(\boldsymbol{\upsilon}_\circ - \boldsymbol{\upsilon}^* + \mathbb{F}^{-1}\boldsymbol{A})\| &\le \frac{\ell^2}{2}\,\tau_3 \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^2 \le \frac{8}{9}\,\tau_3 \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^2. \end{aligned}$$

Also,

$$-\frac{\ell^2(3-\ell)}{6}\,\tau_3 \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^3 \le 2f_\circ(\boldsymbol{\upsilon}_\circ) - 2f_\circ(\boldsymbol{\upsilon}^*) + \|\mathbb{F}^{-1/2}\boldsymbol{A}\|^2 \le \frac{1}{3}\tau_3 \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^3,$$

yielding in particular

$$\bigl|2f_\circ(\boldsymbol{\upsilon}_\circ) - 2f_\circ(\boldsymbol{\upsilon}^*) + \|\mathbb{F}^{-1/2}\boldsymbol{A}\|^2\bigr| \le \frac{1}{2}\,\tau_3 \|D\,\mathbb{F}^{-1}\boldsymbol{A}\|^3.$$

A.5 Quadratic penalization

Here we discuss the case when $f_\circ(\boldsymbol{\upsilon}) - f(\boldsymbol{\upsilon})$ is quadratic. The general case can be reduced to the situation with $f_\circ(\boldsymbol{\upsilon}) = f(\boldsymbol{\upsilon}) - \|G\boldsymbol{\upsilon}\|^2/2$. To make the dependence of G more explicit, denote $f_G(\boldsymbol{\upsilon}) \stackrel{\text{def}}{=} f(\boldsymbol{\upsilon}) - \|G\boldsymbol{\upsilon}\|^2/2$,

$$\boldsymbol{\upsilon}^* = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon}), \quad \boldsymbol{\upsilon}_G^* = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f_G(\boldsymbol{\upsilon}) = \operatorname*{arginf}_{\boldsymbol{\upsilon}} \bigl\{f(\boldsymbol{\upsilon}) - \|G\boldsymbol{\upsilon}\|^2/2\bigr\}.$$

We study the bias $\boldsymbol{\upsilon}_G^* - \boldsymbol{\upsilon}^*$ induced by this penalization.

Define

$$\mathbb{F}_G \stackrel{\text{def}}{=} \nabla^2 f_G(\boldsymbol{\upsilon}^*), \qquad \mathtt{b}_G \stackrel{\text{def}}{=} \| D\, \mathbb{F}_G^{-1} G^2 \boldsymbol{\upsilon}^* \| \,. \tag{A.6}$$

Theorem A.4. *Let*

- *the minimizer* $\boldsymbol{\upsilon}^* = \operatorname{arginf}_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon})$ *exists and unique;*
- $f_G(\boldsymbol{\upsilon})$ *be strongly convex and* $\boldsymbol{\upsilon}_G^* = \operatorname{arginf}_{\boldsymbol{\upsilon}} f_G(\boldsymbol{\upsilon})$ *;*
- $f(\boldsymbol{\upsilon})$ *satisfy* $(\boldsymbol{\mathcal{T}_3^*})$ *at* $\boldsymbol{\upsilon}^*$ *with some* D^2 , τ_3 , *and* $\mathtt{r} = \mathtt{r}(\boldsymbol{\upsilon}^*)$ *;*
- *for* $\mathtt{b}_G = \| D\, \mathbb{F}_G^{-1} G^2 \boldsymbol{\upsilon}^* \|$ *and* $\mathbb{F}_G = \nabla^2 f(\boldsymbol{\upsilon}^*) + G^2$ *; see* (A.6)*:*

$$D^2 \le \varkappa^2 \mathbb{F}_G \,, \qquad \mathtt{r} \ge \frac{4}{3} \mathtt{b}_G \,, \qquad \varkappa^2 \tau_3 \, \mathtt{b}_G \le \frac{3}{8} \,.$$

Then

$$\begin{aligned} \| D(\boldsymbol{\upsilon}_G^* - \boldsymbol{\upsilon}^*) \| &\le \frac{4}{3} \, \mathtt{b}_G \,, \\ \bigl\| D^{-1} \mathbb{F}_G (\boldsymbol{\upsilon}_G^* - \boldsymbol{\upsilon}^* + \mathbb{F}_G^{-1} G^2 \boldsymbol{\upsilon}^*) \bigr\| &\le \frac{8}{9} \, \tau_3 \, \mathtt{b}_G^2 \,, \\ \Bigl| 2 f_G(\boldsymbol{\upsilon}_G^*) - 2 f_G(\boldsymbol{\upsilon}^*) + \| \mathbb{F}_G^{-1/2} G^2 \boldsymbol{\upsilon}^* \|^2 \Bigr| &\le \frac{1}{2} \, \tau_3 \, \mathtt{b}_G^3 \,. \end{aligned} \tag{A.7}$$

A.6 A generic perturbation

This section presents a generic perturbation lemma for (A.5) without any assumption on the structure of the perturbation $f_\circ(\boldsymbol{\upsilon}) - f(\boldsymbol{\upsilon})$. Let $f_\circ(\boldsymbol{\upsilon})$ be a perturbed version of $f(\boldsymbol{\upsilon})$. We consider in parallel two optimization problems

$$\boldsymbol{\upsilon}^* = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon}), \qquad \boldsymbol{\upsilon}_\circ = \operatorname*{arginf}_{\boldsymbol{\upsilon}} f_\circ(\boldsymbol{\upsilon}) \,.$$

The goal is to describe the displacement $\boldsymbol{\upsilon}_\circ - \boldsymbol{\upsilon}^*$ induced by this perturbation. The statement of Theorem A.3 and its proof can be easily extended to this situation. Define

$$\begin{aligned} \mathbb{F} &\stackrel{\text{def}}{=} \nabla^2 f(\boldsymbol{\upsilon}^*) \,, & \mathbb{F}_\circ &\stackrel{\text{def}}{=} \nabla^2 f_\circ(\boldsymbol{\upsilon}^*), \\ b &\stackrel{\text{def}}{=} \| D\, \mathbb{F}^{-1} \nabla f_\circ(\boldsymbol{\upsilon}^*) \| \,, & b_\circ &\stackrel{\text{def}}{=} \| D\, \mathbb{F}_\circ^{-1} \nabla f_\circ(\boldsymbol{\upsilon}^*) \| \,. \end{aligned} \tag{A.8}$$

Theorem A.5. *Let*

- *the minimizer* $\boldsymbol{\upsilon}^* = \operatorname{arginf}_{\boldsymbol{\upsilon}} f(\boldsymbol{\upsilon})$ *exist and unique;*
- $f_\circ(\boldsymbol{\upsilon})$ *be strongly convex and* $\boldsymbol{\upsilon}_\circ = \operatorname{arginf}_{\boldsymbol{\upsilon}} f_\circ(\boldsymbol{\upsilon})$ *;*

- $f_\circ(\upsilon)$ *satisfy* $(\mathcal{T}_3^*)$ *at* υ^* *with some* D^2, τ_3, *and* $\mathtt{r} = \mathtt{r}(\upsilon^*)$;
- *for* $\mathbb{F}_\circ = \nabla^2 f_\circ(\upsilon^*)$, $b_\circ = \|D\,\mathbb{F}_\circ^{-1}\nabla f_\circ(\upsilon^*)\|$ *from* (A.8), *and some* $\varkappa_\circ > 0$

$$D^2 \le \varkappa_\circ^2 \mathbb{F}_\circ\,, \qquad \mathtt{r} \ge \frac{4}{3} b_\circ\,, \qquad \varkappa_\circ^2 \tau_3\, b_\circ \le \frac{3}{8}\,.$$

Then

$$\begin{aligned} \|D(\upsilon_\circ - \upsilon^*)\| &\le \frac{4}{3} b_\circ\,, \\ \bigl\|D^{-1}\mathbb{F}_\circ(\upsilon_\circ - \upsilon^* + \mathbb{F}_\circ^{-1}\nabla f_\circ(\upsilon^*))\bigr\| &\le \frac{8}{9}\,\tau_3\, b_\circ^2\,, \\ \Bigl|2f_\circ(\upsilon_\circ) - 2f_\circ(\upsilon^*) + \|\mathbb{F}_\circ^{-1/2}\nabla f_\circ(\upsilon^*)\|^2\Bigr| &\le \frac{1}{2}\,\tau_3\, b_\circ^3\,. \end{aligned}$$

In some settings, using the Hessian $\mathbb{F}_\circ$ is inconvenient. The next result restates the third-order bounds with $\mathbb{F}$ in place of $\mathbb{F}_\circ$ under a condition on $\mathbb{F}_\circ - \mathbb{F}$.

Theorem A.6. *Let*

- $f(\upsilon)$ *be strongly convex and* $\upsilon^* = \operatorname{arginf}_\upsilon f(\upsilon)$, $\mathbb{F} = \nabla^2 f(\upsilon^*)$;
- $f_\circ(\upsilon)$ *be strongly convex and* $\upsilon_\circ = \operatorname{arginf}_\upsilon f_\circ(\upsilon)$, $\mathbb{F}_\circ = \nabla^2 f_\circ(\upsilon^*)$;
- $f_\circ(\upsilon)$ *satisfy* $(\mathcal{T}_3^*)$ *at* υ^* *with some* D^2, τ_3, *and* $\mathtt{r} = \mathtt{r}(\upsilon^*)$;
- *For some* Δ

$$\bigl\|D^{-1}(\mathbb{F}_\circ - \mathbb{F})D^{-1}\bigr\| \le \Delta\,.$$

- *for* $b = \|D\,\mathbb{F}^{-1}\nabla f_\circ(\upsilon^*)\|$ *from* (A.8), *and some* $\varkappa > 0$ *with* $\varkappa^2\Delta < 1$

$$D^2 \le \varkappa^2\,\mathbb{F}\,, \quad \mathtt{r} \ge \frac{4}{3(1-\varkappa^2\Delta)}\,b\,, \quad \frac{\varkappa^2}{(1-\varkappa^2\Delta)^2}\,\tau_3\, b \le \frac{3}{8}\,.$$

Then with $b_* \stackrel{\mathrm{def}}{=} \dfrac{b}{1-\varkappa^2\Delta}$

$$\begin{aligned} \|D(\upsilon_\circ - \upsilon^*)\| &\le \frac{4}{3}\,b_*\,, \\ \bigl\|D^{-1}\mathbb{F}(\upsilon_\circ - \upsilon^* + \mathbb{F}^{-1}\nabla f_\circ(\upsilon^*))\bigr\| &\le \frac{8}{9}\,\tau_3\, b_*^2 + \frac{4}{3}\,\Delta\, b_*\,, \\ \Bigl|2f_\circ(\upsilon_\circ) - 2f_\circ(\upsilon^*) + \|\mathbb{F}^{-1/2}\nabla f_\circ(\upsilon^*)\|^2\Bigr| &\le \frac{1}{2}\,\tau_3\, b_*^3 + \Delta\, b_*^2\,. \end{aligned}$$

B Application

In the application, we plan to apply the proposed IRMF approach to analyze use the DroneRF dataset, a publicly available collection of radio-frequency (RF) recordings developed for drone detection and identification. The dataset contains recordings associated with three drone types—Bebop, AR, and Phantom—as well as background recordings collected without an active drone. The signals were acquired in a laboratory setting using two NI-USRP 2943R receivers operating in the 2.4 GHz band. The receivers captured complementary lower and upper frequency sub-bands, providing a combined bandwidth of 80 MHz. The complete dataset comprises 227 recording segments stored in 454 CSV files, with each segment represented by a paired set of lower- and upper-band files (DroneRF repository).

We restrict our analysis to distinguish Bebop and AR drones. This restriction defines a binary classification task in which the objective is to distinguish between two drone types under some common operating condition.

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